ONLINE INTERNSHIP INFORMATION PORTAL

AI23431 – WEB TECHNOLOGY AND MOBILE APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

VANDANA M (2116241801306)

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

RAJALAKSHMI ENGINEERING COLLEGE (AUTONOMOUS),

CHENNAI – 602 105

APRIL 2026

RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled **"ONLINE INTERNSHIP INFORMATION PORTAL"** is the Bonafide work of VANDANA M (2116241801306) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Submitted to Project Viva-Voce Examination held on _______________

INTERNAL EXAMINER	EXTERNAL EXAMINER

ACKNOWLEDGEMENT

Initially I thank the Almighty for being with me through every walk of life and showering blessings through the endeavor to put forth this report. My sincere thanks to our Chairman **Mr. S**. **MEGANATHAN, B.E, F.I.E.,** our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN**, **B.E., M.S.,** and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.,** for providing the requisite infrastructure and sincere endeavoring in educating students in their premier institution.

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VANDANA M – 2116241801306

DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

- PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.
- PEO 2: To prepare students with fundamental knowledge in programming languages, tools and enable them to develop applications.
- PEO 3: To encourage the research abilities and innovative project development in the field of AI, Data Science, networking, security, web development and also emerging technologies for the cause of social benefit.
- PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry requirements.

PROGRAMME OUTCOMES (POs)

- PO 1: Engineering Knowledge – Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

- PO 2: Problem Analysis – Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

- PO 3: Design/Development of Solutions – Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

- PO 4: Conduct Investigations of Complex Problems – Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

- PO 5: Modern Tool Usage – Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

- PO 6: The Engineer and Society – Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

- PO 7: Environment and Sustainability – Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

- PO 8: Ethics – Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

- PO 9: Individual and Team Work – Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

- PO 10: Communication – Communicate effectively on complex engineering activities with the engineering community and with society at large.

- PO 11: Project Management and Finance – Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

- PO 12: Life-long Learning – Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Data Science Program will demonstrate:

- PSO 1: Foundation Skills – Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, artificial intelligence, data science, machine learning, and networking for solving real-world problems.

- PSO 2: Problem-Solving Skills – Ability to apply mathematical methodologies to solve computational tasks, model real-world problems using appropriate AI and data science algorithms, and deliver quality products using open-ended programming environments.

- PSO 3: Successful Progression – Ability to apply knowledge in various domains to identify research gaps and provide solutions to new ideas, inculcate passion towards higher studies, creating innovative career paths and evolving as an ethically responsible AI and Data Science professional.

COURSE OBJECTIVES

- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML5, CSS3, and JavaScript, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming using Vanilla JavaScript.
- To implement client-side storage using Firebase Firestore as a real-time cloud data persistence layer.To facilitate students in understanding multi-page web application design with role-based navigation and dynamic content rendering.
- To help students gain understanding of third-party API integration including Firebase Authentication with real-time Phone OTP verification.

COURSE OUTCOMES

On completion of the course, students will be able to:

- CO 1: Design and develop structured, semantic, multi-page web applications with consistent navigation and responsive layouts using HTML5 and CSS3.
- CO 2: Implement client-side dynamic behaviour including form validation, modal dialogs, real-time content filtering, and interactive internship cards using Vanilla JavaScript.
- CO 3: Apply Firebase Firestore as a persistent cloud database for storing and retrieving students, companies, internship listings, and application data in real time.
- CO 4: Integrate Firebase Authentication for secure user registration and login with phone number OTP-based verification.
- CO 5: Design and implement a role-based access control system with distinct student, company, and admin workflows, including an internship approval pipeline, LMS learning system, and admin management panel.

CO-PO-PSO MAPPING

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO 1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO 2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO 3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO 4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO 5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low)   2: Moderate (Medium)   3: Substantial (High)   No correlation: "–"

SUSTAINABLE DEVELOPMENT GOALS

SDG GOAL	GOAL	JUSTIFICATION
SDG-4	Quality Education	InternHub's integrated LMS Academy promotes quality culinary education by enabling students to access professional skill-building courses online, supporting lifelong learning for all academic levels.
SDG-8	Decent Work and Economic Growth	The platform empowers students to gain industry exposure and companies to find skilled interns, supporting economic opportunity through a structured digital hiring portal.
SDG-9	Industry, Innovation, and Infrastructure	InternHub uses innovative web technologies (Firebase OTP, Firestore database, LMS, dynamic approval workflows) to build accessible digital infrastructure for internship knowledge sharing.
SDG-17	Partnerships for the Goals	The multi-role portal fosters collaboration between students, companies, and admin coordinators, integrating third-party services like Firebase, reflecting partnerships for shared digital career goals.

ABSTRACT

InternHub is a community-driven Online Internship Information Portal developed as a university mini-project using HTML5, CSS3, and Vanilla JavaScript, backed by Google Firebase as the cloud database and authentication provider. The system delivers a fully functional multi-page web application where students can register using a phone-number-based OTP login, browse internship listings posted by companies, apply for internships, and track their application status in real time.

The application is built around three distinct user roles. Students access a comprehensive Student Dashboard where they can search and filter internship listings, submit applications, monitor their application pipeline (pending, shortlisted, offered, rejected), and access a built-in Learning Management System (LMS) to take courses and earn certificates. Companies access a Company Dashboard where they can post internship listings, review incoming applications, shortlist candidates, and send offer letters. All company-posted internships are stored in Firebase Firestore under a pending status and are made publicly visible only after an administrator verifies and approves them.

The administrator role is protected by a dedicated Admin Authentication page and grants access to the Admin Control Panel. This panel provides complete control over company registrations (approve/reject), internship post approvals, student and company account management, platform-wide analytics and reports, and activity logs. The Firebase Firestore backend stores all user accounts, company profiles, internship documents, and application records in structured collections.

The visual design employs a stunning dark-themed interface with gradient backgrounds, animated elements, and a clean card-based layout for an immersive and professional user experience. Navigation is streamlined through a login page with role selection that directs users to their respective dashboards. The project successfully demonstrates all core web technologies covered in the AI23431 course curriculum and validates a complete, real-world internship portal workflow.

TABLE OF CONTENTS

LIST OF ABBREVIATIONS

Abbreviation	Full Form
IIP	Online Internship Information Portal
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
OTP	One-Time Password
API	Application Programming Interface
UI	User Interface
LMS	Learning Management System
CRUD	Create, Read, Update, Delete
Firebase	Google Backend-as-a-Service Platform
Firestore	Firebase Cloud Firestore Database
CDN	Content Delivery Network
SDG	Sustainable Development Goal
AI	Artificial Intelligence
DS	Data Science
HTTP	HyperText Transfer Protocol
URL	Uniform Resource Locator
RBAC	Role-Based Access Control

CHAPTER 1

INTRODUCTION

1.1 General

Internship programs have traditionally been managed through college placement cells, physical notice boards, and word-of-mouth networks. In the digital era, the internet has transformed how internships are discovered, applied for, and managed. However, most existing internship portals lack structured quality control for company verification, an integrated learning system for students, and a simple, role-based interface for all three stakeholders — students, companies, and administrators.

InternHub addresses this gap by providing a community-driven Online Internship Information Portal where company-posted internships are subject to administrator review before publication. Built using HTML5, CSS3, and Vanilla JavaScript on the frontend and Google Firebase on the backend, the system demonstrates a complete, real-world internship management workflow suited for a university-level web technology project.

1.2 Need for InternHub

Existing internship platforms suffer from three core problems. First, they lack quality control — any company can post any internship without verification, resulting in fraudulent or misleading listings reaching students. Second, they require complex server-side infrastructure beyond the scope of most university projects. Third, there is no single platform that integrates internship listing, application tracking, LMS-based skill building, and role-based dashboards for all three user types — students, companies, and admins — using only standard web technologies.

InternHub solves this by introducing an admin-controlled company verification and internship approval pipeline. Students can apply for verified internships, track their applications in real time, and upskill through integrated LMS courses. This ensures a consistently high standard of content and a safe, trustworthy experience for all users.

1.3 Scope of the System

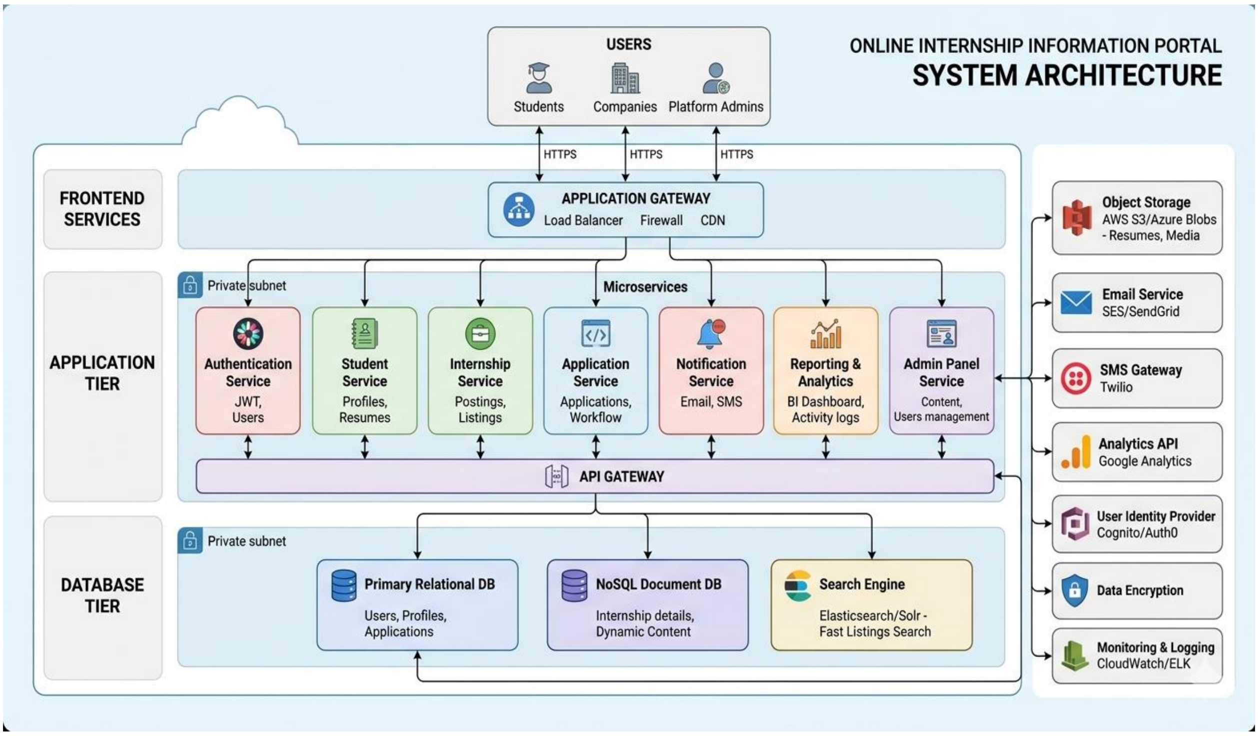
The scope of InternHub encompasses seven core functional areas: student and company registration with Firebase OTP authentication; internship listing and search with filters; a student application system with real-time status tracking; a company dashboard for posting internships, reviewing applications, and sending offer letters; an admin panel for approving companies and internship posts, managing users, and viewing analytics; a built-in Learning Management System (LMS) with course enrollment and certificate tracking; and a notification system for all role-based updates.

1.4 Organization of the Report

This report is organized into seven chapters:

- Chapter 1: Provides the introduction, need, scope, and organization of the project.
- Chapter 2: Presents the literature survey discussing existing culinary platforms and related web technology research.
- Chapter 3: Explains the problem formulation and specific objectives of FlavorShare.
- Chapter 4: Describes the system design including architecture, data model, and requirements.
- Chapter 5: Discusses the full implementation of all modules with screenshots.
- Chapter 6: Presents the results, performance evaluation, and system testing.
- Chapter 7: Concludes the project and discusses possible future enhancements.

1.5 System Architecture Diagram



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The intersection of academic career management and digital technology has generated significant research around online job portals, community-driven internship platforms, and cloud-based web applications. Traditional internship discovery began with static notice boards and has evolved into dynamic, real-time platforms with database integration, multimedia content, and role-based access control features. This chapter reviews existing systems, identifies their limitations, and positions InternHub within the current landscape of web-based career and internship management applications.

2.2 Existing Internship and Job Platforms

Platforms such as Internshala, LinkedIn, and Naukri represent the current state of online internship and job portals. These platforms allow company submissions and student applications but lack a structured admin verification workflow at the university level, meaning unverified or low-quality internship listings frequently reach students. Furthermore, none of these platforms are built as client-side-only applications — they rely on complex server-side infrastructures inaccessible for university-level projects.

LinkedIn and Indeed support job applications but are not purpose-built for internship-specific workflows — they lack college-specific filters, integrated LMS modules, and a structured admin approval pipeline. Internshala provides internship listings but has no admin verification at the university coordinator level and no built-in learning management system.

2.3 Related Work

Research by Naaman et al. (2010) on community content curation demonstrates that editorial gatekeeping significantly improves perceived content quality in online communities. Their findings directly support the admin approval model at the core of InternHub. Studies on Firebase as a web application backend by Google Engineering (2023) show that Firestore's real-time document model is well-suited for content management systems with moderate traffic, making it ideal for a university project deployment.

Work on role-based access control (RBAC) in web applications demonstrates that separating student, company, and admin interfaces significantly reduces unauthorized content manipulation — a design principle implemented through InternHub's separate login flows and role-specific dashboards.

2.4 Summary

The literature survey reveals that while many internship platforms exist, none combine a structured Firebase Firestore backend, admin approval workflow, integrated LMS, application tracking system, and a clean three-role interface in a single client-side web application built solely with HTML5, CSS3, and Vanilla JavaScript. InternHub occupies a unique niche by delivering all of these features without any server-side language, making it particularly relevant as a university project.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

Existing online internship platforms suffer from three core problems. First, they lack quality control — any company can post any internship without verification, resulting in fraudulent or misleading listings reaching students. Second, they require complex server-side infrastructure (Node.js, PHP, databases) that is beyond the scope of most university projects. Third, there is no simple demonstration of a real Firebase-backed web application that covers student, company, and admin workflows in a single cohesive project using only standard web technologies.

There is a clear need for a well-structured, Firebase-backed internship portal that demonstrates user authentication, Firestore database integration, role-based navigation, an integrated LMS, application tracking, and a content moderation pipeline — all built with HTML5, CSS3, and Vanilla JavaScript.

3.2 Objectives of the System

The specific objectives of InternHub are:

- To implement a secure user registration and login system using Firebase Authentication with phone number OTP-based verification for students and companies.
- To build an internship listing interface where companies can post internship details including role, stipend, duration, location, eligibility, and required skills.
- To store all internship postings in Firebase Firestore with a pending status, ensuring no company listing is publicly visible before admin verification.
- To develop a Student Dashboard with internship search and filter, application submission, real-time application status tracking (pending, shortlisted, offered, rejected), saved internships, and profile management.
- To develop a Company Dashboard enabling companies to post internships, review incoming student applications, shortlist candidates, and send offer letters.
- To develop an Admin Control Panel with three management areas — Company Approvals, Internship Post Approvals, and User Management — along with platform-wide analytics and activity logs.
- To integrate a built-in Learning Management System (LMS) where students can enroll in courses, track progress, and earn certificates to strengthen their internship profile.
- To deliver a visually immersive interface using a dark-themed gradient design with animated elements, a clean card-based layout, and a consistent colour palette.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design translates the project objectives into a concrete technical architecture. InternHub is designed as a multi-page client-side application where each HTML page handles a specific role, and all data operations are performed directly against the Firebase Firestore database and Firebase Authentication service via the Firebase JavaScript SDK. The design emphasises clean separation of concerns: page structure in HTML, visual styling in CSS, and all dynamic behaviour and Firebase interactions in JavaScript.

4.2 System Architecture

InternHub follows a three-layer client-side architecture:

- Presentation Layer: Seven HTML pages — login.html (OTP-based login with role selection), registration.html (student and company signup), student-dashboard.html (student hub), company-dashboard.html (company hub), admin-dashboard.html (admin control panel), internship-listings.html (public internship browse), and lms.html (Learning Management System).
- Logic Layer: Inline JavaScript within each page handles Firebase SDK calls, DOM manipulation, form validation, modal dialogs, and navigation routing. All Firebase reads and writes use Firestore's real-time document API.
- Data Layer: Google Firebase Firestore stores four collections — users (students and companies), internships (role, company, stipend, status), applications (student, internship, status, timestamp), and courses (LMS data). Firebase Authentication manages OTP-based session state.

4.3 Data Model Design

The Firebase Firestore data model uses the following collection structures:

- users collection: Each document stores { name, phone, role (student/company/admin), college, degree, branch, cgpa } for students, or { companyName, hrName, email, industry, location } for companies.
- internships collection: Each document stores { company, role, stipend, duration, location, mode, skills, description, status (pending/approved/active/closed), deadline, timestamp } — the full internship data along with its current moderation status.

- applications collection: Each document stores { studentId, internshipId, company, role, coverLetter, resumeURL, status (pending/shortlisted/offered/rejected), appliedOn } — tracking every application through its lifecycle.
- courses collection: Each document stores { title, category, instructor, duration, enrolled, progress, certificate } — the LMS course data for student learning tracking.

4.4 System Requirements

4.4.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or equivalent / Any ARM processor
RAM	Minimum 2 GB (4 GB recommended)
Storage	Minimum 50 MB for project files; Firebase stores data in cloud
Display	Any screen resolution 1024×768 or above
Internet	Required for Firebase Firestore, Firebase Authentication, and CDN resources

4.4.2 Software Requirements

Component	Specification
Operating System	Windows 10+, macOS 10.14+, Ubuntu 20.04+
Browser	Google Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
Frontend Technologies	HTML5, CSS3, Vanilla JavaScript (ES6+)
Backend / Database	Google Firebase — Firestore (database) + Authentication
Firebase SDK	Firebase JS SDK v9+ (modular) via CDN
Development IDE	Visual Studio Code (recommended)
Internet Connection	Required at runtime for all Firebase operations

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

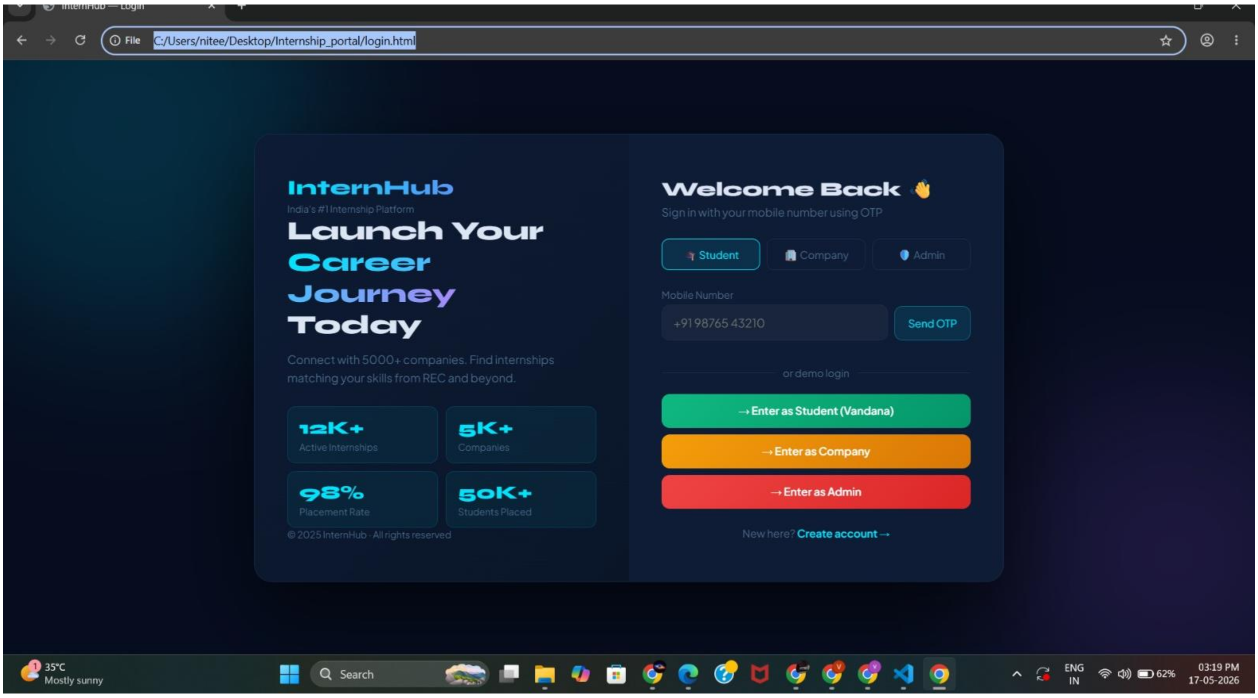
InternHub is implemented using a page-per-role architecture. Development began with the Firebase project setup — creating the Firestore database with appropriate security rules, and enabling Firebase Authentication. The frontend pages were then built in sequence: the login page with OTP, student registration, company registration, student dashboard, company dashboard, admin dashboard, internship listings, and LMS. Each page imports the Firebase SDK via CDN and initialises the Firebase app with the project configuration object.

The visual design uses a dark-themed gradient background with animated floating particles and glowing orbs. All interactive elements — buttons, input fields, navigation tabs, and cards — use a consistent blue-cyan accent colour palette. JavaScript handles all Firestore reads, writes, and real-time updates without any server-side code.

5.2 Module Descriptions

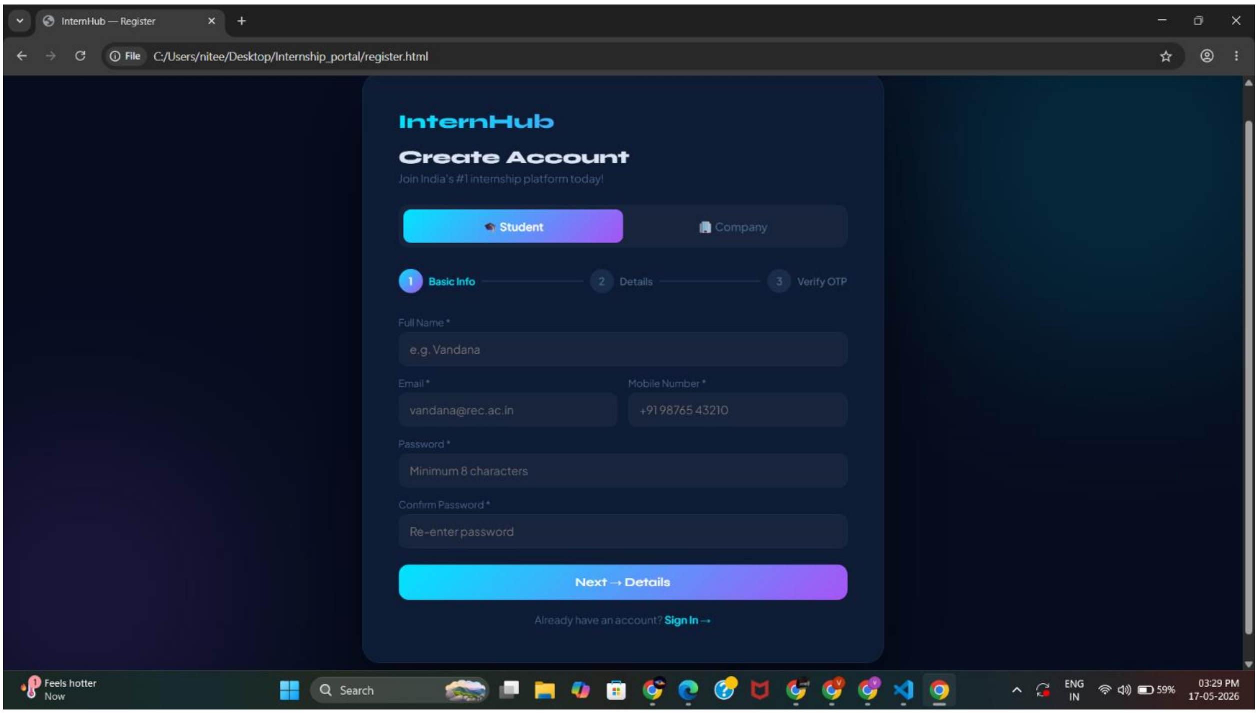
5.2.1 Login / OTP Authentication Module (login.html)

The login page serves as the entry point of InternHub. It displays the InternHub brand name over an animated dark background with floating particles and glowing gradient orbs. Three role selector pills — Student, Company, and Admin — allow the user to select their role before logging in. The user enters their mobile number and clicks Send OTP, which triggers the Firebase Authentication phone sign-in flow. A six-box OTP entry grid appears, and upon entering the correct OTP, the user is automatically redirected to their role-specific dashboard. A bottom navigation bar links all pages for easy demonstration during the project review.



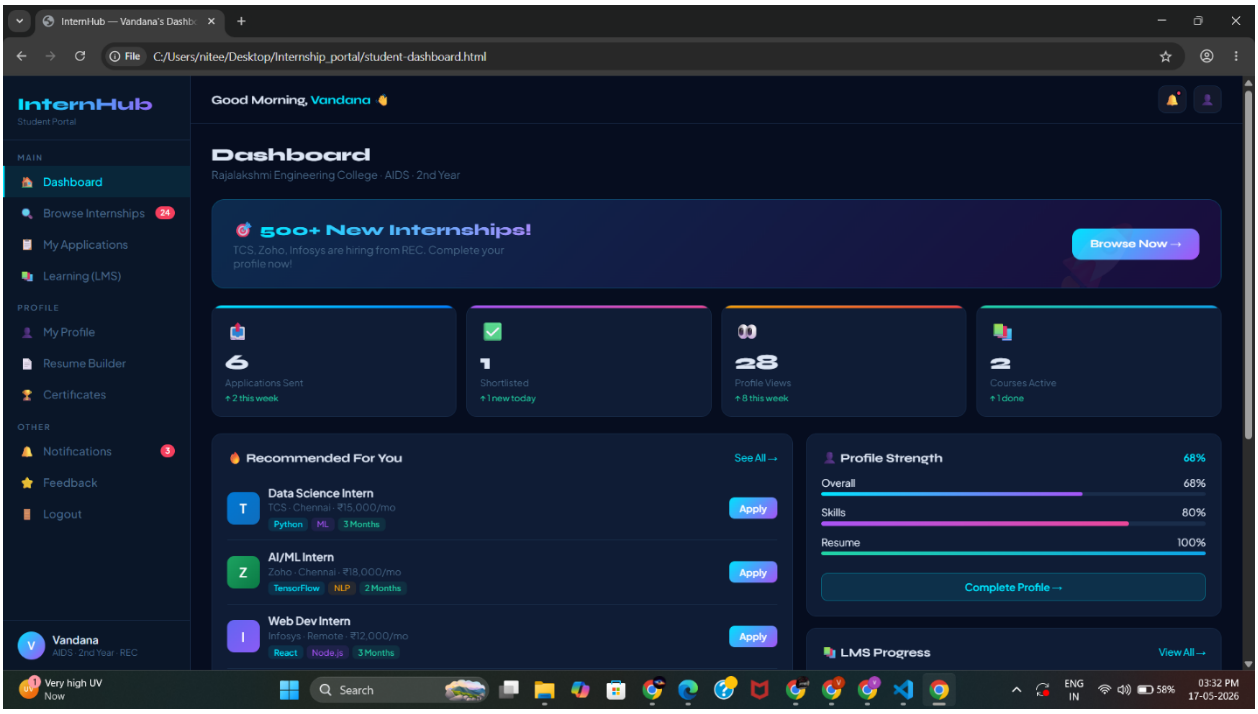
5.2.2 Registration Module (registration.html)

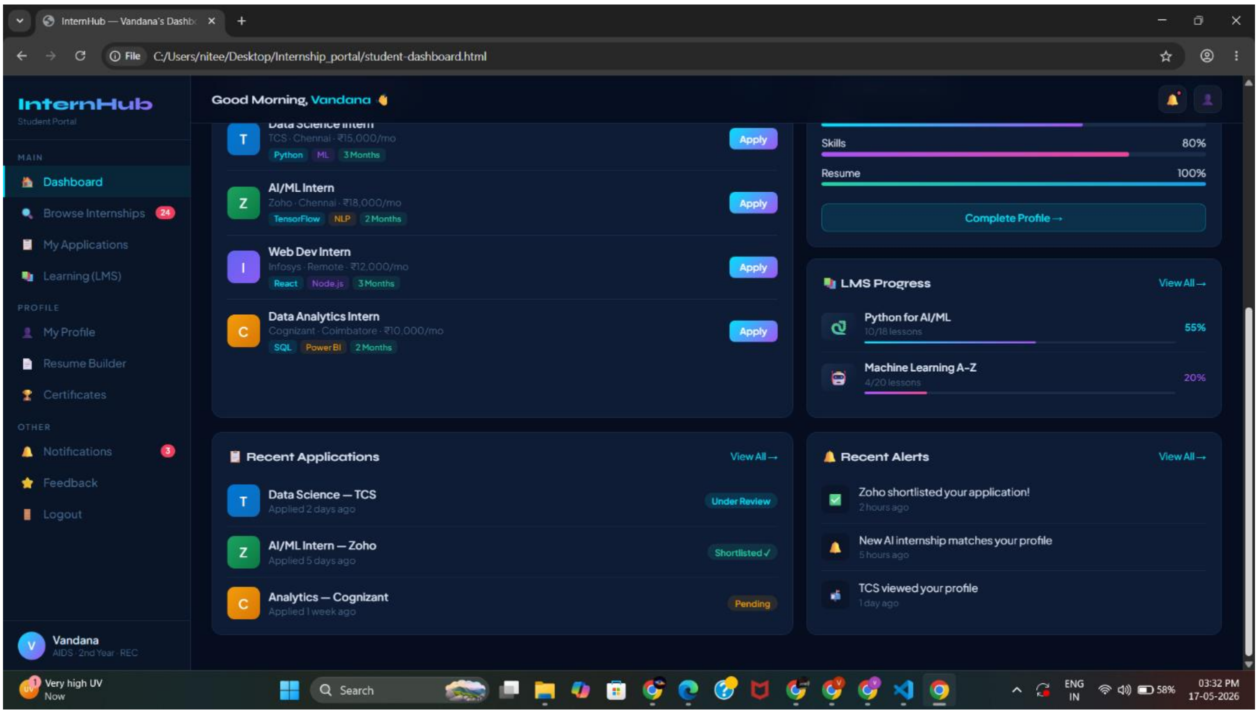
The registration page allows new users to create an account. Two tab options — Student and Company — switch the form fields dynamically. Student registration collects name, mobile, email, college, degree, branch, year, and CGPA. Company registration collects company name, HR name, designation, email, phone, industry, location, and website. Upon submission, the user document is written to the Firestore users collection and the user is redirected to their respective dashboard. The page maintains a consistent dark theme with gradient card styling.



5.2.3 Student Dashboard Module (student-dashboard.html)

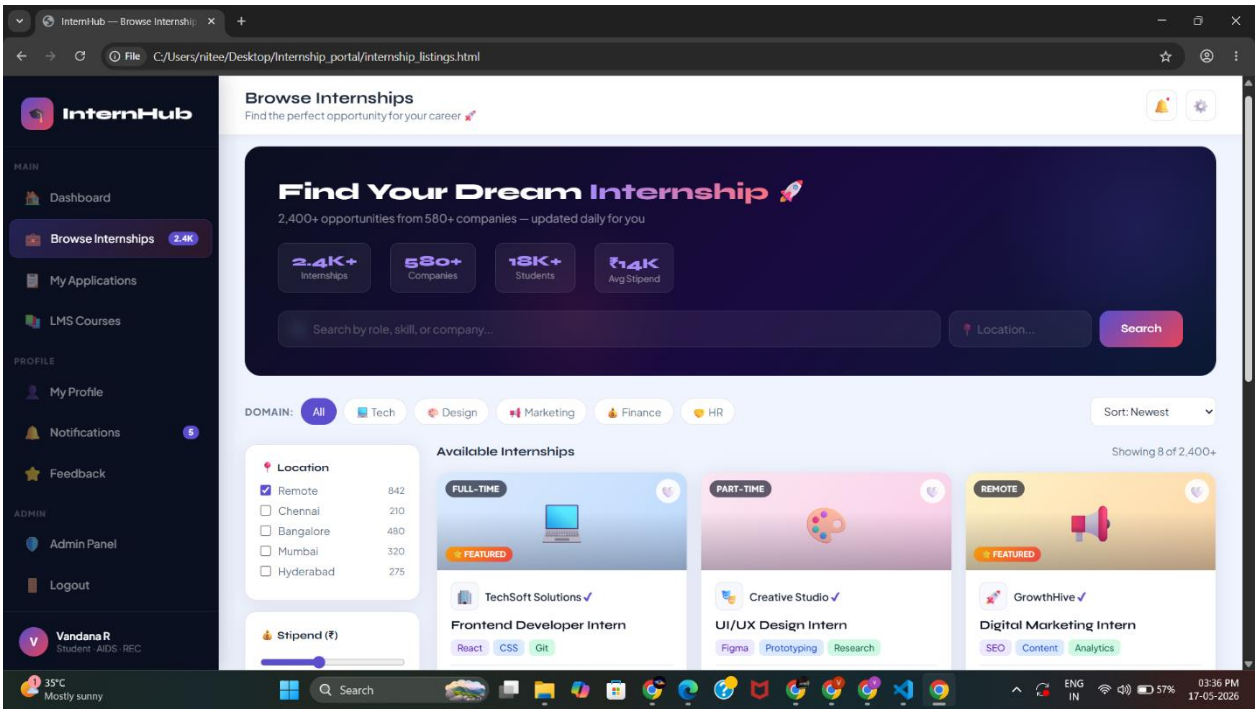
The Student Dashboard is the main hub for student users. A fixed sidebar provides navigation to all student features: Overview (stats and quick access), Find Internships, My Applications, Saved Internships, LMS Courses, Certificates, My Profile, Notifications, and Settings. The Overview page displays four stat cards (Applications Sent, Shortlisted, Saved Internships, Profile Strength), a recent applications table with status badges (Pending, Shortlisted, Offered, Rejected), and upcoming application deadlines. A welcome banner with a call-to-action encourages students to browse internships.





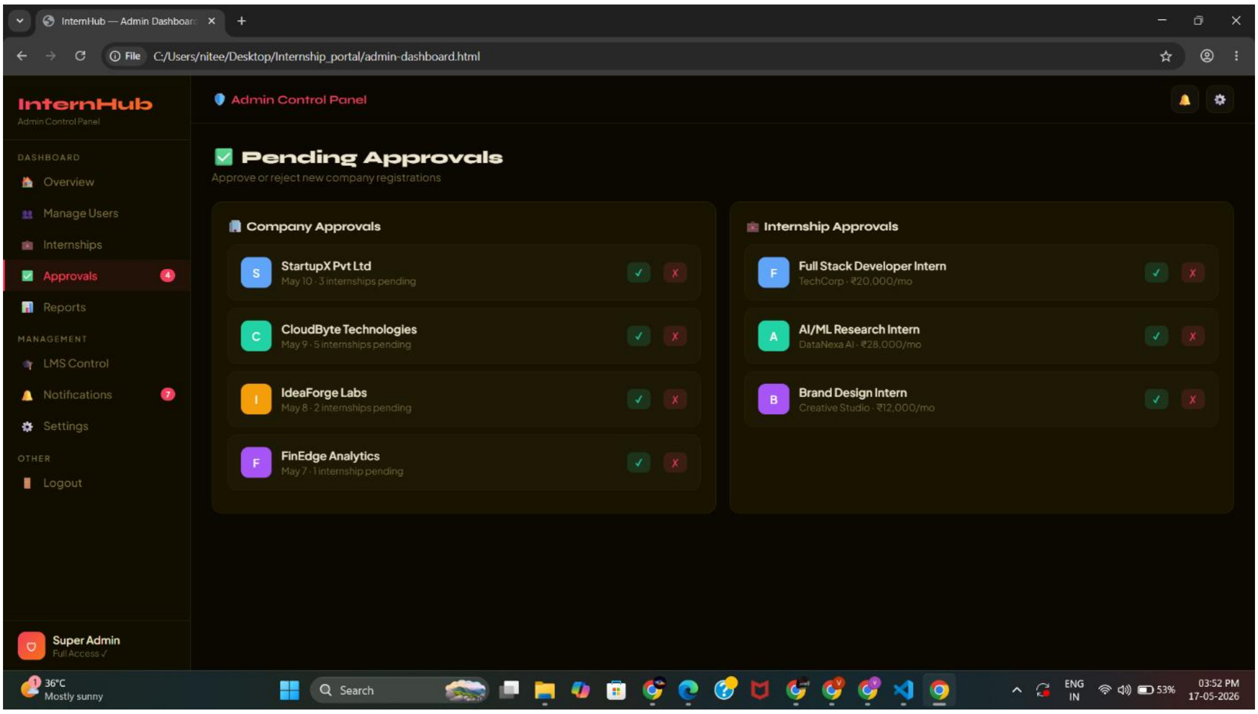
5.2.4 Internship Listings Module (internship-listings.html)

The internship listings page provides a searchable, filterable grid of all approved internship postings. A top search bar allows keyword search by role, company, or skill. Filter chips below allow quick filtering by domain (IT, Data Science, Design, Marketing, Finance), work mode (Remote, Hybrid), and stipend range. Each internship card displays the company logo emoji, company name, role title, skill tags, stipend, deadline, and an Apply Now button. Clicking Apply Now opens a modal form for cover letter submission and resume upload, which writes an application document to the Firestore applications collection.



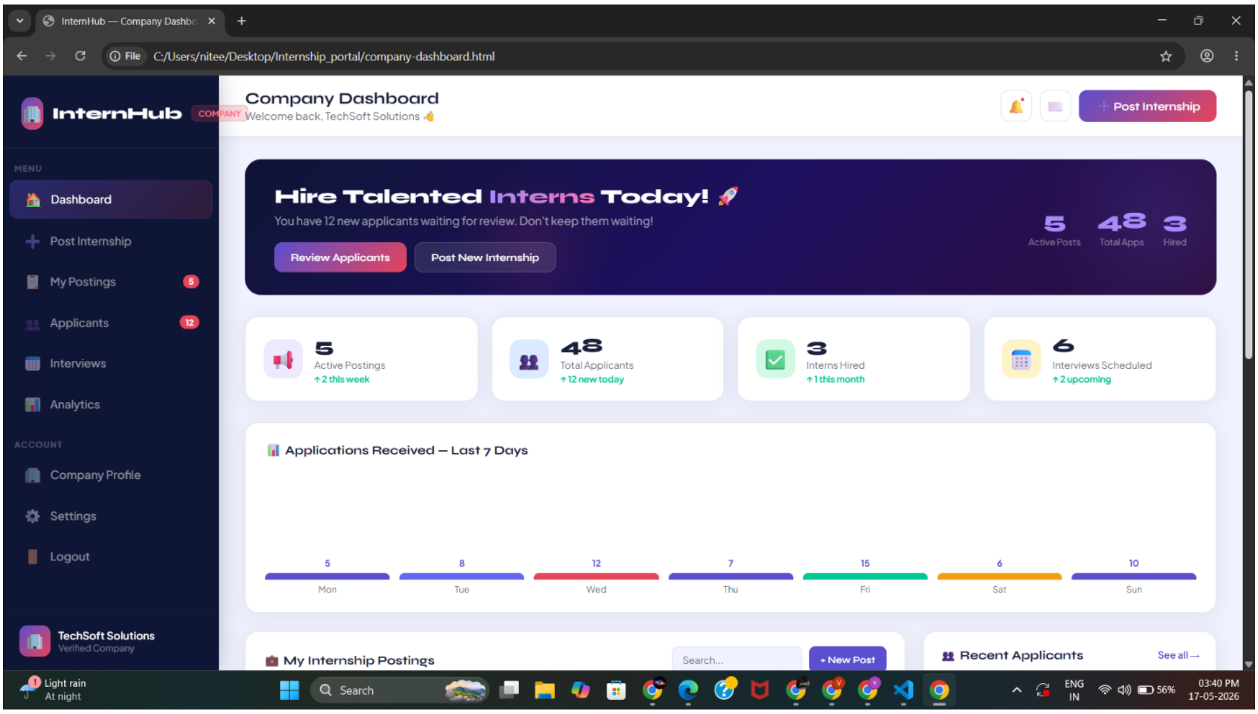
5.2.5 Application Tracking Module (application-tracking.html)

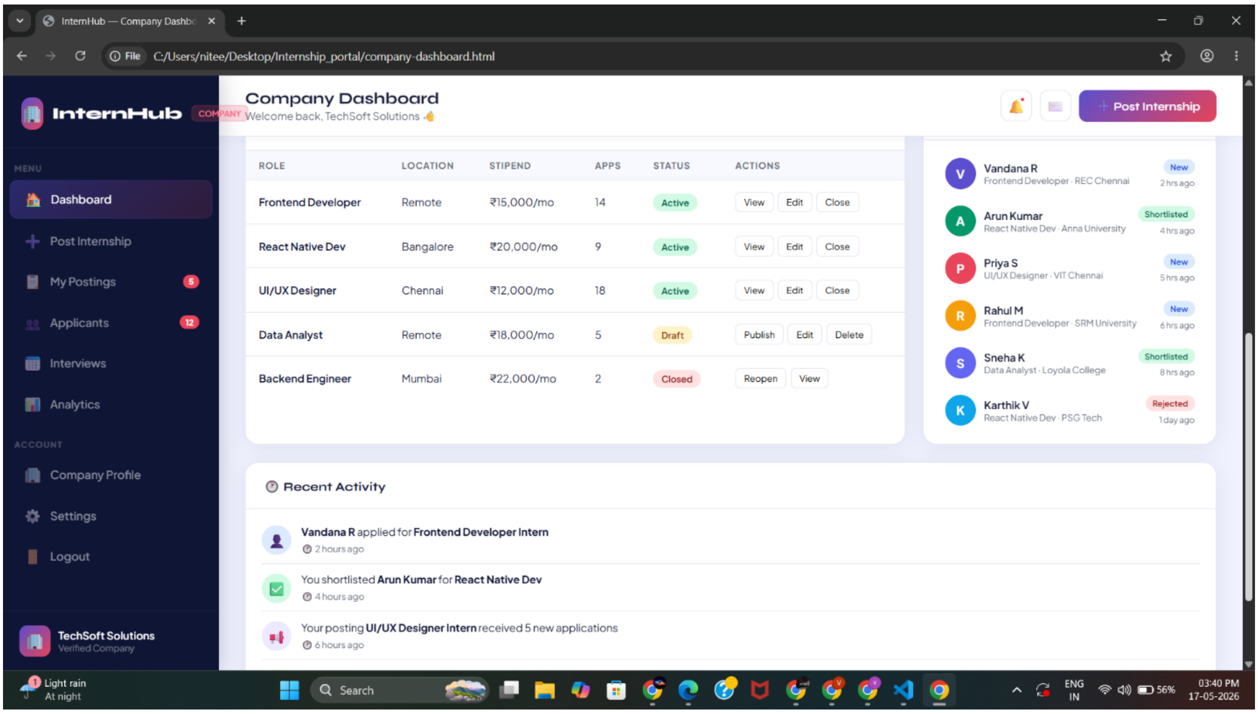
The application tracking page presents a comprehensive table of all the logged-in student's internship applications. Each row shows company name, role, duration, stipend, date applied, and a colour-coded status badge. Status values include Pending (yellow), Shortlisted (green), Offered (blue), and Rejected (red). Students can view application details, accept offers, or request feedback on rejections directly from this page. The status data is queried from the Firestore applications collection filtered by the logged-in student's ID.



5.2.6 Company Dashboard Module (company-dashboard.html)

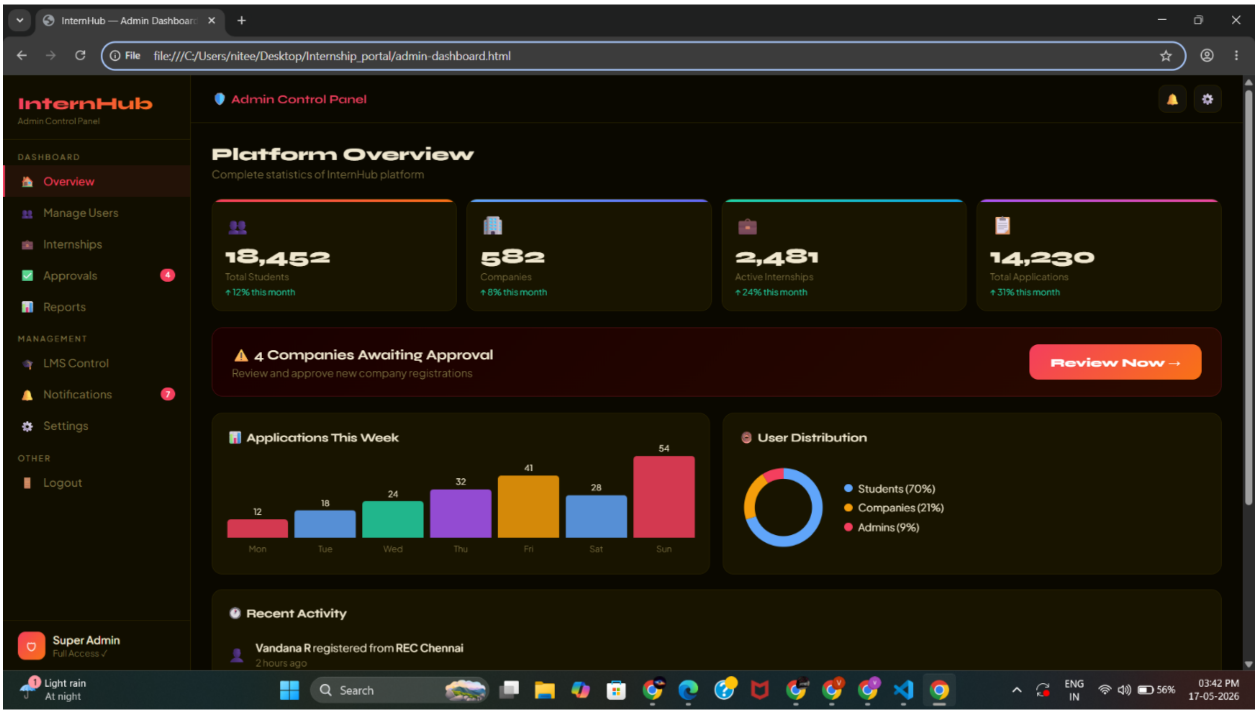
The Company Dashboard provides the hiring interface for company users. A fixed dark sidebar navigates to: Dashboard Overview, Post Internship, My Listings, All Applicants, Shortlisted, Offers Sent, Analytics, Notifications, and Company Profile. The overview page displays four stat cards (Active Listings, Total Applicants, Shortlisted, Offers Sent), a latest applicants table with Shortlist, Resume, and Reject action buttons, a quick stats bar chart showing application distribution by role, and a recent activity feed. The Post Internship form collects all internship details and writes to Firestore with a pending status awaiting admin approval.

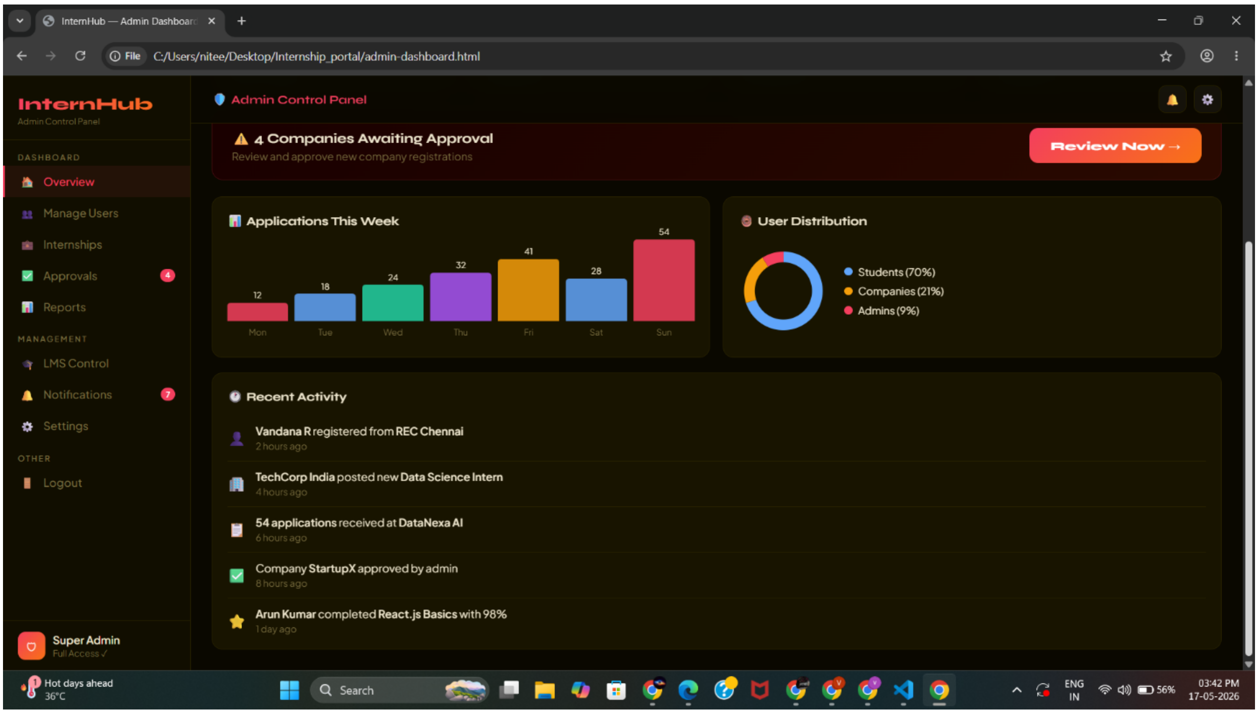




5.2.7 Admin Dashboard Module (admin-dashboard.html)

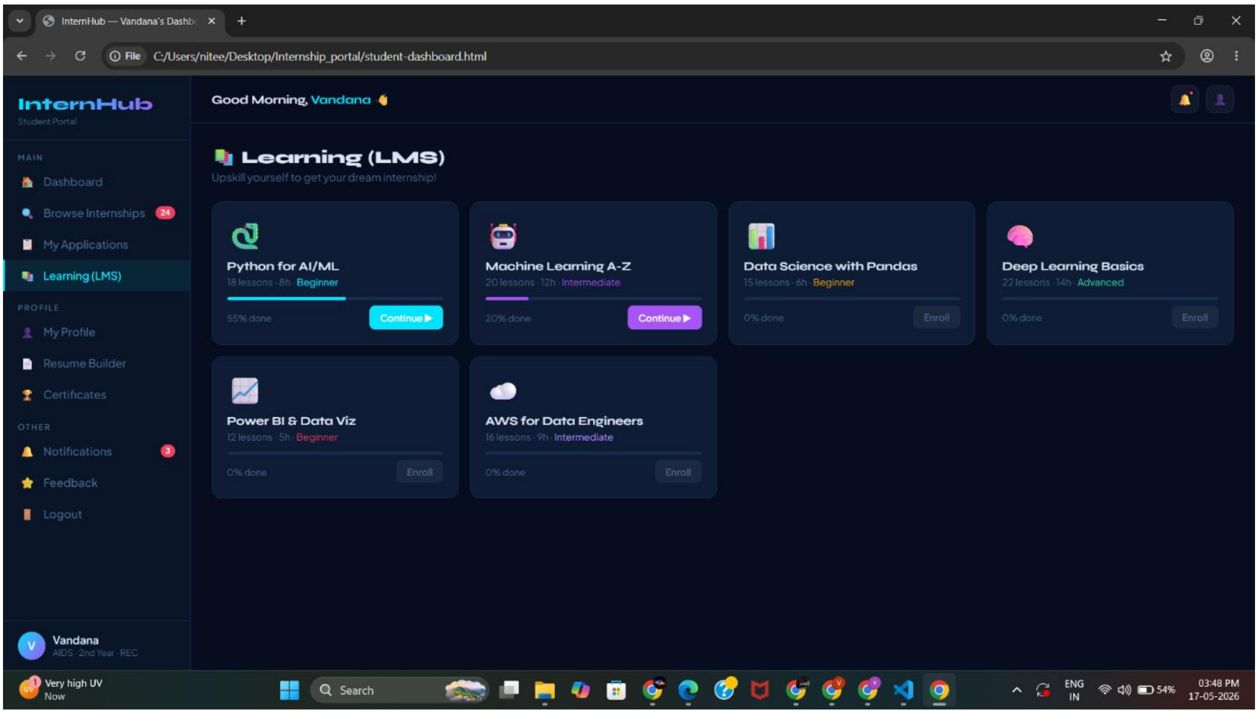
The Admin Control Panel provides complete oversight of the InternHub platform. A dark sidebar with orange accent colour navigates to: Dashboard Overview, Reports and Analytics, Company Approvals, Internship Post Approvals, All Students, All Companies, All Internships, Reviews, Notifications, Settings, and Activity Logs. The overview page displays platform-wide stats (Total Students, Registered Companies, Active Internships, Placements This Month), a pending companies approval table with Approve and Reject buttons, a recent alerts feed, and quick conversion funnel charts. All admin actions update Firestore document statuses in real time.





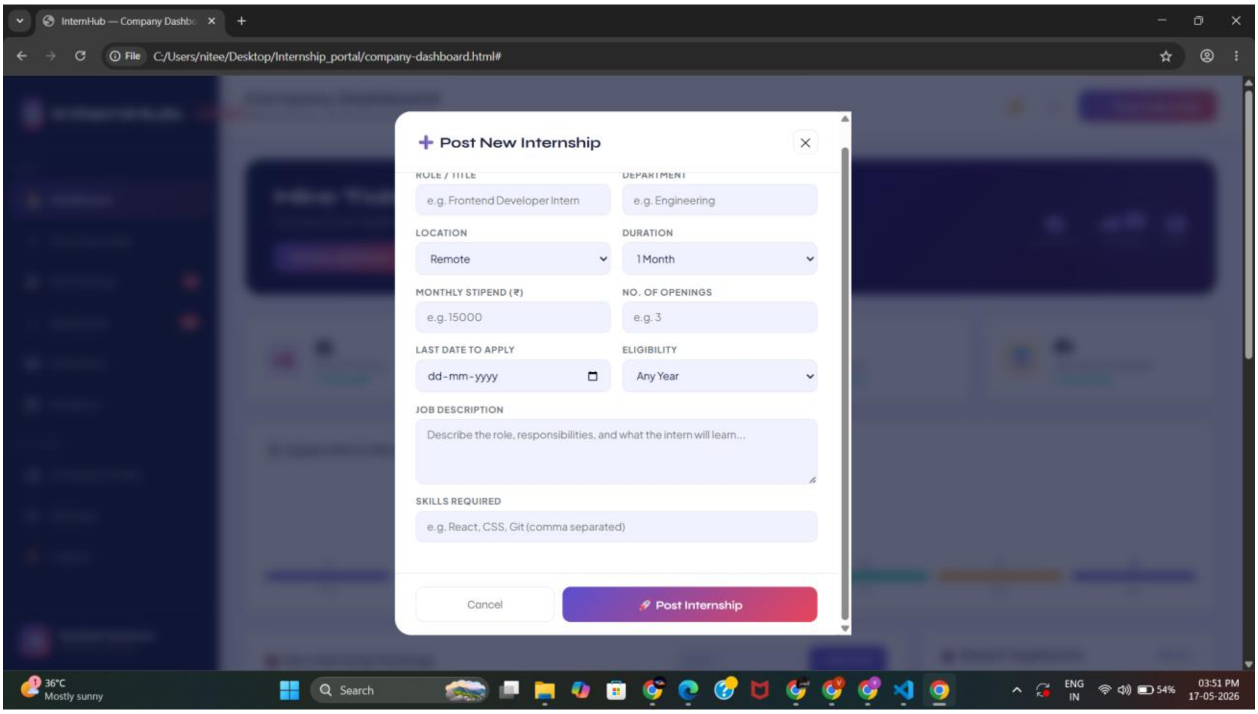
5.2.8 LMS Module (lms.html)

The Learning Management System module provides students with access to professional skill-building courses. A hero banner displays enrollment stats (courses enrolled, completed, certificates earned). Below, a responsive card grid displays available courses with a thumbnail gradient, category label, course title, instructor name, duration, rating, and a progress bar showing the student's current completion percentage. Completed courses display a green Certificate Earned badge. The Certificates section allows students to download earned certificates as PDF and upload externally earned certificates to strengthen their profile.



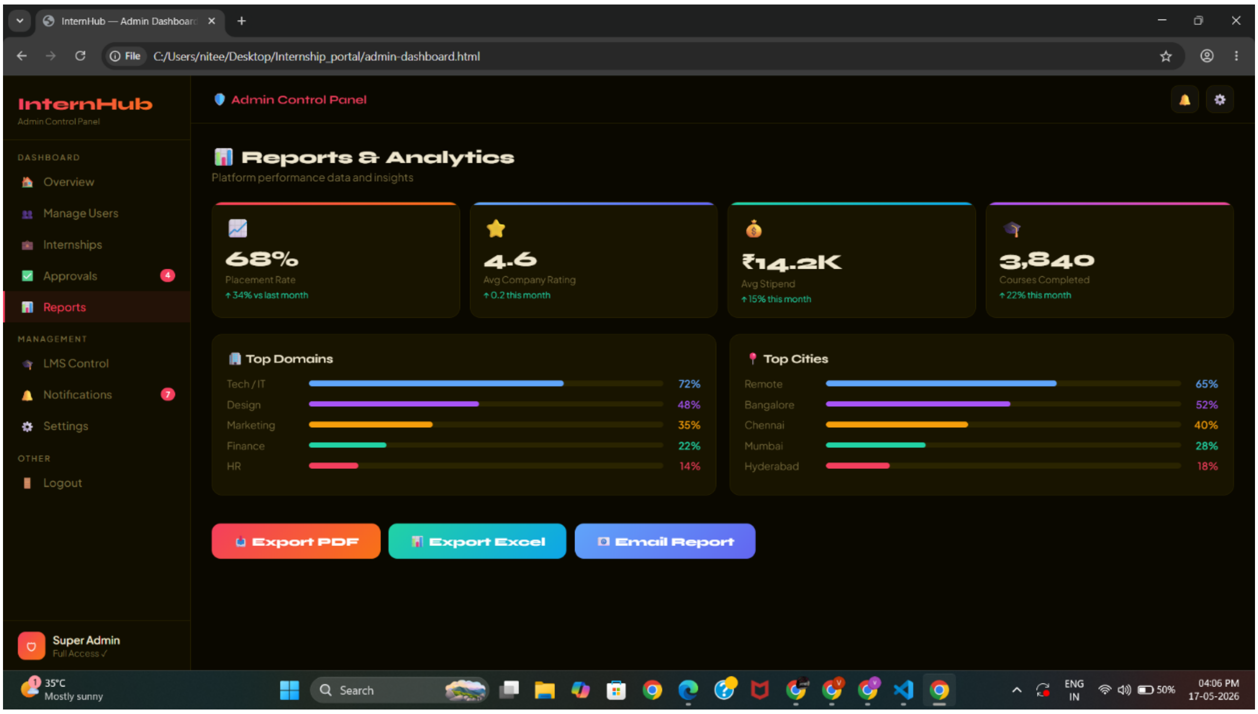
5.2.9 Firebase Firestore — Data Collections

The Firebase Firestore database stores four structured collections. The users collection stores all registered student and company documents with their respective fields. The internships collection stores all company-posted internship documents with a status field (pending, approved, active, closed) that controls visibility. The applications collection stores all student applications with a status field (pending, shortlisted, offered, rejected) that drives the application tracking dashboard. The courses collection stores LMS course data including progress and certificate status for each enrolled student.



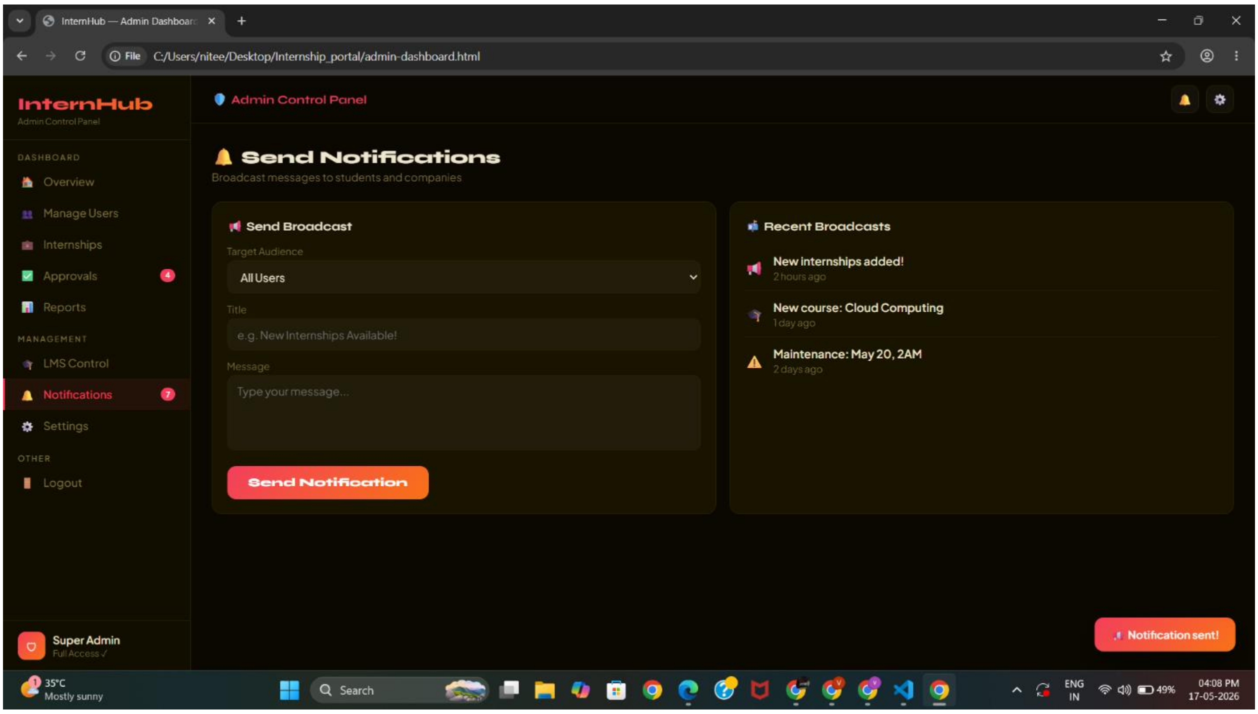
5.2.10 Reports and analaytics module (admin.html)

The **Reports and Analytics Module** is an important part of an **Online Internship Information Portal**. It helps administrators, colleges, companies, and students analyze internship-related data in an organized and meaningful way. This module converts raw data into useful reports, charts, and statistics that support better decision-making and improve the overall efficiency of the portal.



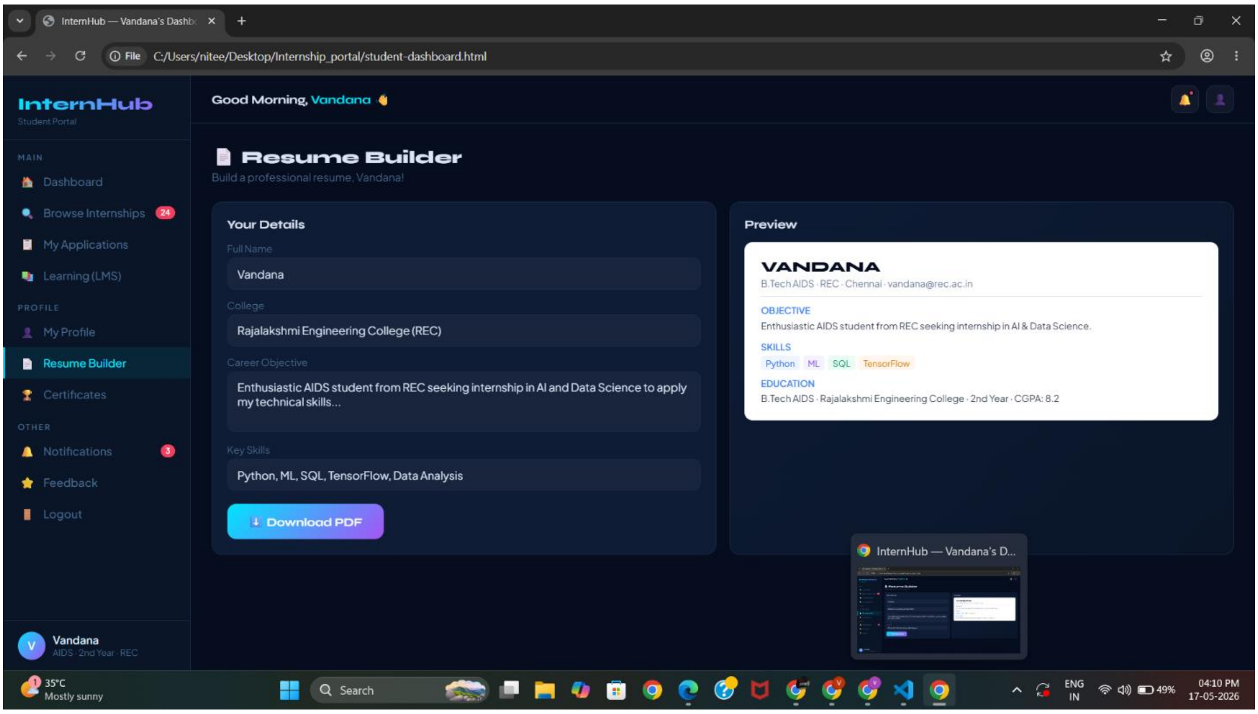
5.2.11 Admin control panel notifications (admin.html)

The Admin Control Panel Notifications Module is an essential component of the Online Internship Information Portal. This module helps administrators receive real-time updates and alerts regarding important activities happening within the system. It improves communication, monitoring, and management efficiency by ensuring that admins are informed about every significant event in the portal.



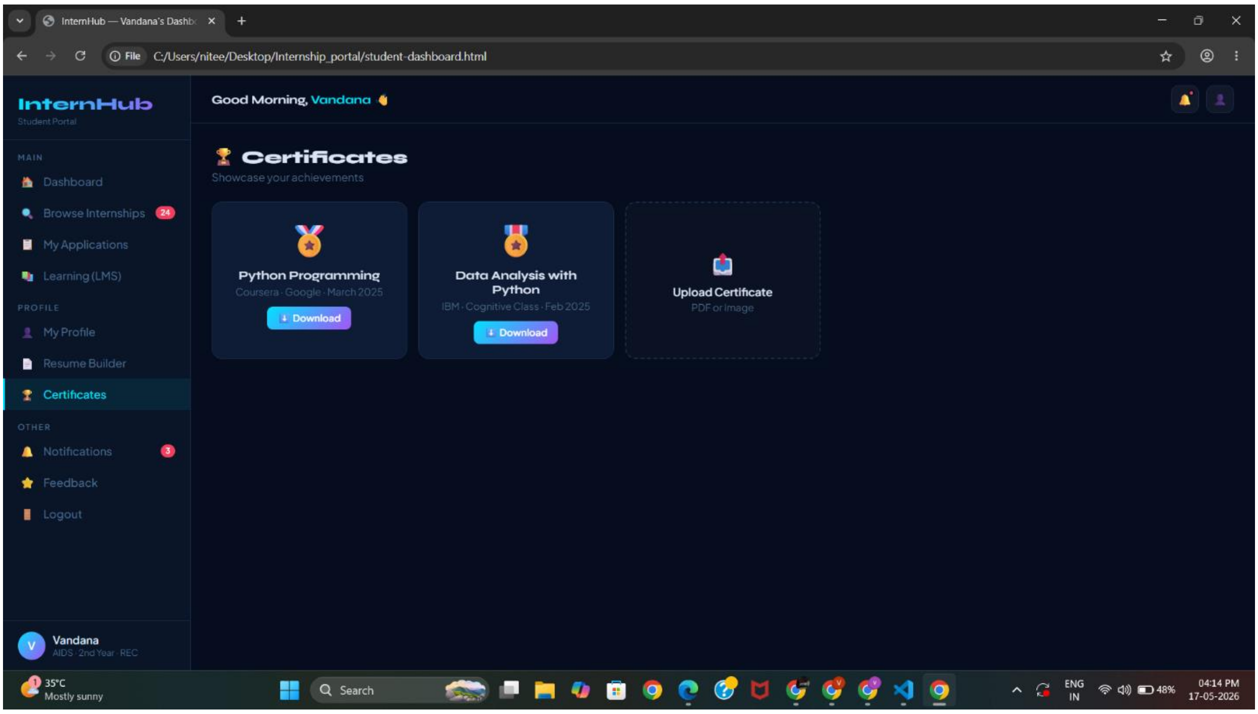
5.2.12 Resume builder (admin.html)

The Resume Builder Module is an important feature of the Online Internship Information Portal. It helps students create professional and well-structured resumes easily without requiring advanced design or formatting skills. This module allows users to enter their academic, personal, and technical details and automatically generates a resume in a professional format suitable for internship applications.



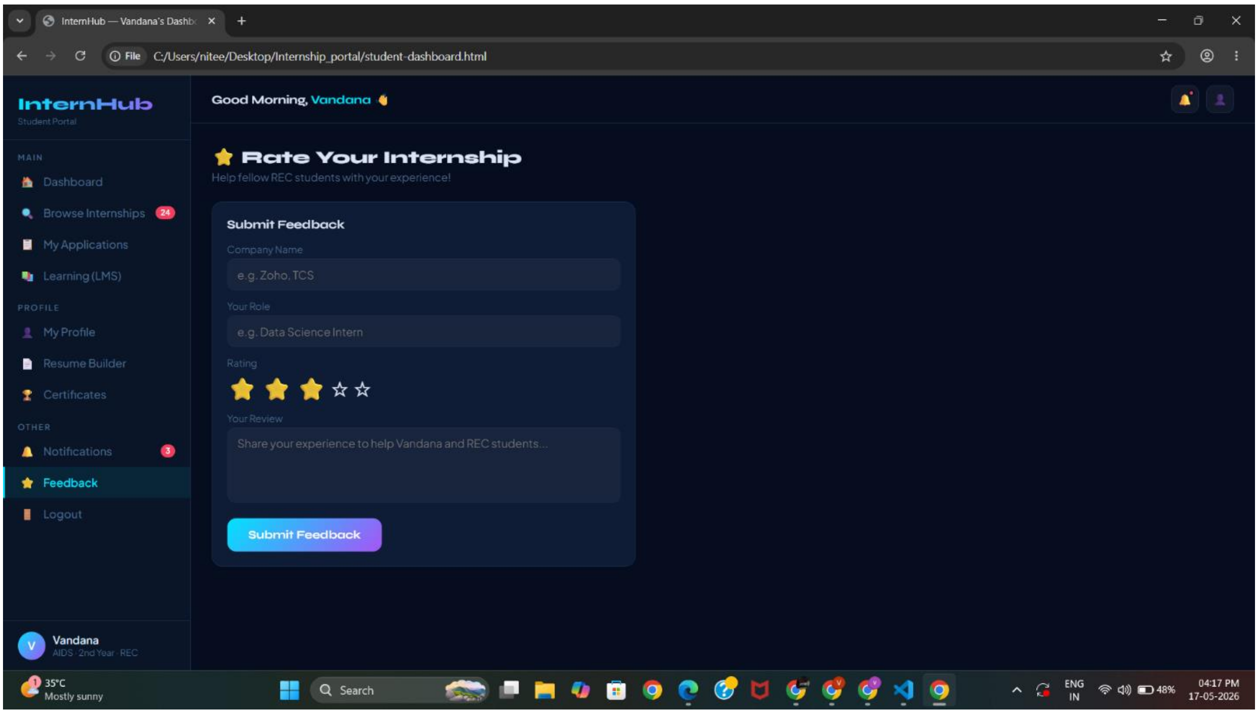
5.2.13 Certifications module (dashboard.html)

This module helps students upload, manage, and showcase their certificates earned from internships, online courses, workshops, seminars, and skill development programs. It acts as a digital repository where users can securely store and access their certifications anytime.



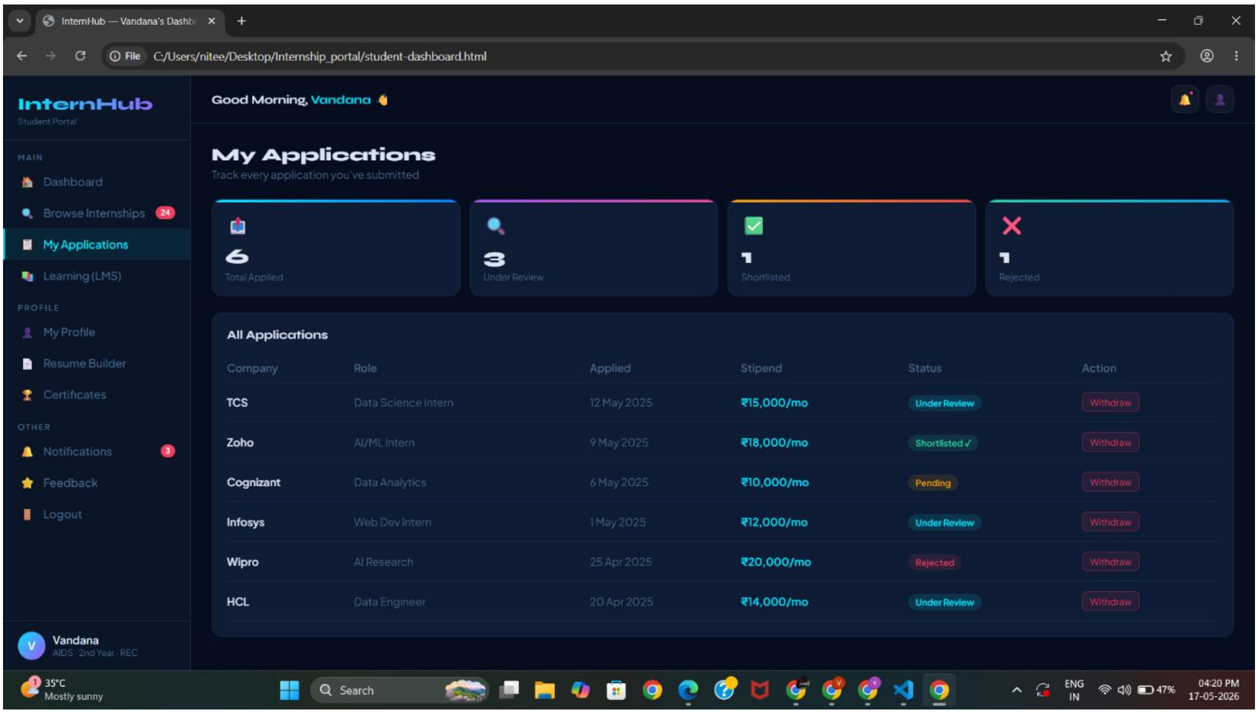
5.2.14 Rating internship module (internship_listings.html)

The Internship Rating and Feedback Module is an important feature of the Online Internship Information Portal. This module allows students to provide ratings and feedback about their internship experiences, companies, mentors, and work environments. It helps future applicants make informed decisions and enables organizations to improve the quality of internship programs.



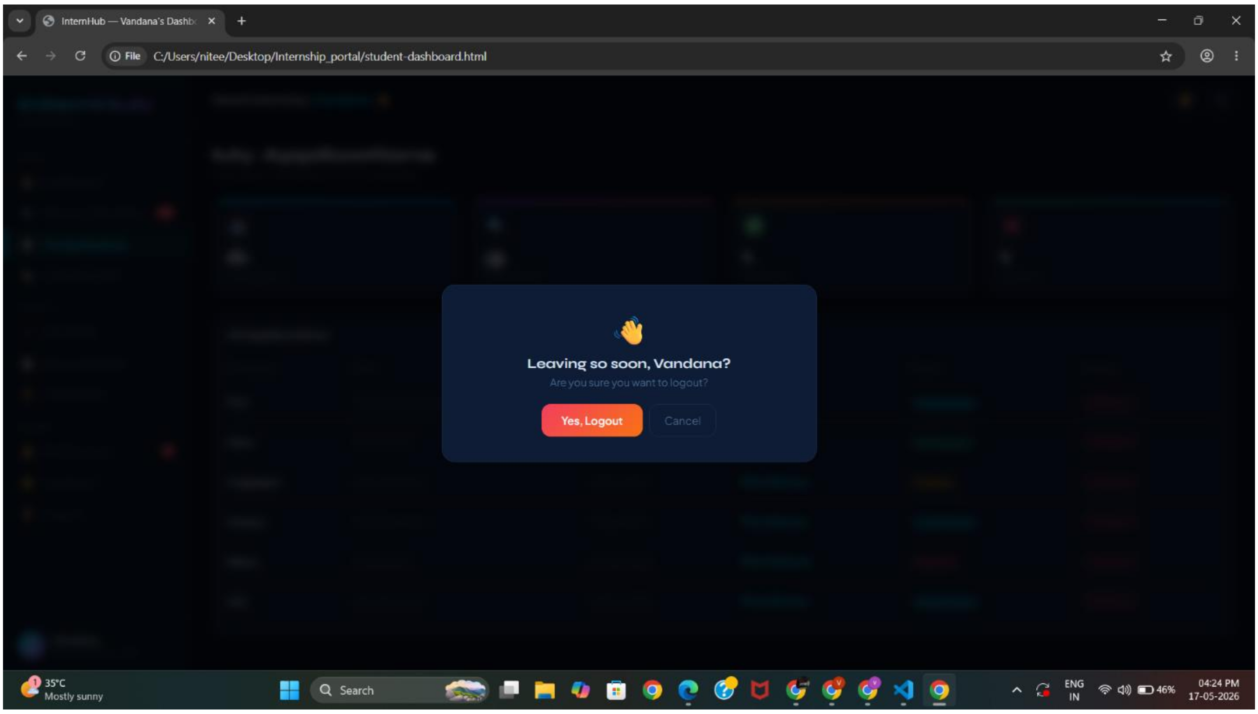
5.2.15 Tracking applications (internship_listings.html)

The Application Tracking Module is a key component of the Online Internship Information Portal. This module helps students, companies, and administrators monitor and manage internship applications efficiently. It provides real-time updates about the status of applications and ensures smooth communication between all users involved in the internship process.



5.2.16 Logout portal

The Logout Module is an important security feature of the Online Internship Information Portal. This module allows users such as students, companies, and administrators to safely exit the system after completing their activities. It helps protect user accounts and sensitive information from unauthorized access.



CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The InternHub Online Internship Information Portal was successfully developed and tested across Chrome, Firefox, and Edge on Windows. All seven pages — login with OTP, registration, student dashboard, internship listings, application tracking, company dashboard, and admin dashboard — rendered correctly and functioned as designed. The Firebase Firestore database correctly stored user, internship, and application documents, and the admin approval workflow was verified end-to-end.

The complete user journey was demonstrated: a new student registered, browsed and applied for an internship, the company reviewed and shortlisted the application, the admin approved the company's listing, and the student received an offer notification. The Firestore console confirmed that all data was correctly structured. Rejected applications appeared in the student's application tracker with the appropriate badge.

6.2 Performance Evaluation

Since InternHub uses Firebase as its backend, performance depends on Firebase SDK response times and the client's internet connection. The following observations were recorded during testing:

- Firebase Firestore document reads (fetching internship listings): approximately 300–600 milliseconds on standard broadband.
- Firebase Firestore document writes (application submission): under 500 milliseconds with confirmation callback.
- Page load time (with Firebase SDK via CDN): approximately 1.5 seconds on first load; under 500 ms on subsequent loads with browser caching.
- Admin panel internship list render (10 listings): imperceptible delay with Firestore's real-time query model.
- OTP delivery (Firebase phone auth): 3–8 seconds depending on network conditions.

The system performed reliably throughout all 14 test cases with no errors or timeouts encountered.

6.3 System Testing

Systematic test cases were designed to cover all major user flows and edge cases. The following table summarises the results:

TC#	Test Case	Input / Action	Expected Output	Result
1	Student Sign-Up	Enter name, phone, OTP → Register	Account created; redirected to Student Dashboard	PASS
2	Student Login	Enter mobile → OTP → Verify	Authenticated; Student Dashboard loads	PASS
3	Company Sign-Up	Enter company details → Register	Account created; redirected to Company Dashboard	PASS
4	Admin Login	Select Admin role → OTP → Verify	Admin session created; Admin Dashboard opens	PASS
5	Internship Post	Fill all fields → Submit to Admin	Internship stored in Firestore with status pending	PASS
6	Admin Approves Company	Admin clicks Approve for company	Company status updated to approved in Firestore	PASS
7	Admin Approves Internship	Admin clicks Approve for listing	Internship becomes active and visible to students	PASS
8	Student Applies	Browse listing → Apply Now → Submit	Application stored in Firestore with status pending	PASS
9	Company Shortlists	HR clicks Shortlist on applicant	Application status updated to shortlisted	PASS
10	Company Sends Offer	HR fills offer modal → Send	Offer letter sent; application status set to offered	PASS
11	Application Tracking	Student views My Applications tab	All applications shown with correct status badges	PASS
12	LMS Enrollment	Student clicks Enroll on course	Course progress tracked; certificate earned on 100%	PASS
13	Firestore Data Verify	Check Firebase console collections	All collections show correct document structure	PASS
14	Logout	Click Logout on any page	Session cleared; redirected to login page	PASS

6.4 Resource Utilisation

InternHub's frontend files total under 150 KB (excluding Firebase SDK loaded via CDN). The Firebase Spark (free) plan was used throughout development and testing, which provides 1 GB of Firestore storage, 50,000 daily reads, 20,000 daily writes, and 20,000 daily deletes — more than sufficient for a university project with a small team of testers. No paid plan is required for course demonstration purposes.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

InternHub successfully demonstrates that a full-featured, Firebase-backed Online Internship Information Portal can be built using only HTML5, CSS3, and Vanilla JavaScript. The project implements every component of a real-world internship management platform — student and company authentication via Firebase OTP, a multi-role dashboard system, an internship posting and approval pipeline, a student application tracking system, an integrated LMS with certificate tracking, Firestore-backed data persistence, and a complete admin control panel — without any server-side programming language or build tools.

The clean visual design, built with a dark-themed gradient interface, animated background effects, and responsive card layouts, delivers a professional and immersive user experience. All eight project objectives stated in Chapter 3 were fully achieved and verified through systematic testing against the live Firebase backend. This project fulfils the requirements of the AI23431 Web Technology and Mobile Application course and provides a practical, deployable demonstration of modern client-side web development with cloud backend integration.

7.2 Future Enhancements

- Real-time Notifications: Implement Firebase Cloud Messaging to send push notifications to students when their application status changes and to companies when new applications arrive.
- Resume Builder: Add an in-portal resume builder that generates a professional PDF resume from the student's profile data stored in Firestore.
- AI-Powered Matching: Integrate an AI recommendation engine that matches student skill profiles with the most relevant internship listings automatically.
- Mobile Application: Develop a React Native companion app using the same Firebase backend, extending InternHub to iOS and Android platforms.
- Video Interview Module: Add an in-portal video interview scheduling and recording feature using WebRTC, allowing companies to conduct initial screening interviews without leaving the platform.
- Analytics Dashboard: Integrate Firebase Analytics to provide administrators and companies with insights into most-applied roles, top-performing colleges, application conversion rates, and placement trends over time.
- College Coordinator Portal: Add a fourth user role for college placement coordinators to monitor their students' application activity and manage campus-specific internship drives.

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APPENDICES

Appendix 1: System Pseudocode

System Overview

InternHub is a client-side web portal backed by Google Firebase. All user accounts, internship listings, and application decisions are stored in Firebase Firestore. Firebase Authentication manages OTP-based session state. No server-side code is used — all Firebase operations are performed directly from JavaScript in the browser.

Student Registration Logic

```
FUNCTION RegisterStudent(name, phone, college, degree, branch, cgpa)
  Validate all fields are non-empty
  Trigger Firebase OTP to phone number
  IF OTP verified THEN
    Create user document in Firestore users collection
    Redirect to Student Dashboard
  END IF
END FUNCTION
```

Application Submission Logic

```
FUNCTION SubmitApplication(studentId, internshipId, coverLetter, resumeURL)
  Validate all fields are non-empty
  Create application document:
    { studentId, internshipId, coverLetter, resumeURL,
      status: 'pending', appliedOn: Date.now() }
  Write document to Firestore applications collection
END FUNCTION
```

Admin Approval Logic

```
FUNCTION ApproveInternship(internshipDocumentId)
  Find internship document in Firestore by ID
  Update document: { status: 'approved' }
END FUNCTION
```

Appendix 2: System Test Verification Summary

- Total test cases executed: 14
- Test cases passed: 14
- Test cases failed: 0
- Firebase Firestore collections verified: users, internships, applications, courses
- Admin approval flow: end-to-end verified with Firestore status update confirmation
- Student application tracking: all four status tabs (pending, shortlisted, offered, rejected) verified
- LMS module: course enrollment, progress tracking, and certificate generation verified

Final System Status: InternHub has been fully developed and successfully tested against a live Firebase Firestore backend. The system ensures secure role-based access via OTP authentication, quality-controlled internship listings through an admin approval workflow, transparent application status tracking for students, a structured hiring pipeline for companies, and integrated skill-building through the LMS module. The professional dark-themed visual interface and Firebase-backed architecture confirm the project's readiness for academic submission and live demonstration.